ARJUNA JEE 2027

Lecture -03 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A Angles & Quadrant

B Question Practice

C

D



Question Practice

Question [JEE Mains-2023]



The sum $\sum_{n=1}^{\infty} \frac{2n^2+3n+4}{(2n)!}$ is equal to

A $\frac{11e}{2}+\frac{7}{2e}$

B $\frac{13e}{4}+\frac{5}{4e}-4$

C $\frac{11e}{2}+\frac{7}{2e}-4$

D $\frac{13e}{4}+\frac{5}{4e}$

$$\frac{2n^2}{(2n)!}+\frac{3n}{(2n)!}+\frac{4}{(2n)!}$$

$2n.(2n-1)!$

$$\frac{2n^2}{2n.(2n-1)!}=\frac{(2n-1)+1}{2.(2n-1)!}=\frac{1}{2}\left(\frac{2n-1}{(2n-1)!}+\frac{1}{(2n-1)!}\right)=\frac{1}{2}\left(\frac{1}{(2n-2)!}+\frac{1}{(2n-1)!}\right)$$

$(2n-1).(2n-2)!$

$$T_n = \frac{1}{2}\left(\frac{1}{(2n-2)!_0} + \frac{1}{(2n-1)!_0}\right) + \frac{3}{2} \times \frac{1}{(2n-1)!_0} + \frac{4}{(2n)!_0}$$

$$T_n = \frac{1}{2} \times \frac{1}{(2n-2)!_0} + 2 \times \frac{1}{(2n-1)!_0} + \frac{4}{(2n)!_0}$$

$$= \frac{1}{2}\left(\frac{1}{0!_0}\right) + 2 \times \frac{1}{1!_0} + 4 \times \frac{1}{2!_0}$$

$$= \frac{1}{2}\left(\frac{1}{2!_0}\right) + 2 \times \frac{1}{3!_0} + 4 \times \frac{1}{4!_0}$$

$$= \frac{1}{2}\left(\frac{1}{4!_0}\right) + 2 \times \frac{1}{5!_0} + 4 \times \frac{1}{6!_0}$$

$$\vdots \qquad \vdots \qquad \vdots$$

$$\frac{1}{0!_0} + \frac{1}{2!_0} + \frac{1}{4!_0} + \ldots\infty = \frac{e^1 + e^{-1}}{2}$$

$$\frac{1}{1!_0} + \frac{1}{3!_0} + \frac{1}{5!_0} + \ldots\infty = \frac{e^1 - e^{-1}}{2}$$

$$= \frac{1}{2}\left(\frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \cdots\right) + 2\times\left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \cdots\right) + 4\times\left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \cdots\right)$$

$$\frac{1}{2}\left(\frac{e^{1}+e^{-1}}{2}\right) + 2\left(\frac{e^{1}-e^{-1}}{2}\right) + 4\left(\frac{e^{1}+e^{-1}}{2} - 1\right)$$

$$\frac{1}{4}\left(e + \frac{1}{e}\right) + e - \frac{1}{e} + 2e + \frac{2}{e} - 4.$$

$$\boxed{\frac{13}{4}e + \frac{5}{4}e^{-1} - 4.}$$

$$\frac{1}{0!} + \frac{1}{2!} + \frac{1}{4!} + \cdots\infty = \frac{e^{1}+e^{-1}}{2}$$

$$\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \cdots\infty = \frac{e^{1}-e^{-1}}{2}$$

Challenger Question [JEE Mains-2021]

If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4 \ldots\ldots$, is equal to

A $x\left(\frac{1+x}{1-x}\right) + \log_e(1-x)$

B $x\left(\frac{1-x}{1+x}\right) + \log_e(1-x)$

C $\frac{1-x}{1+x} + \log_e(1-x)$

D $\frac{1+x}{1-x} + \log_e(1-x)$

$T_n = \frac{(2n+1)\,x^{n+1}}{(n+1)}$ ✓ $\quad n = 1, 2, 3 \ldots$

$= \frac{(2n+2-1)\,x^{n+1}}{(n+1)}$

$= \frac{2(n+1)x^{n+1}}{n+1} - \frac{x^{n+1}}{n+1}$

$T_n = 2x^{n+1} - \frac{x^{n+1}}{n+1}$

$$T_n = 2x^{n+1} - \frac{x^{n+1}}{n+1}$$

$$T_1 = 2x^2 - \frac{x^2}{2}$$

$$T_2 = 2x^3 - \frac{x^3}{3}$$

$$T_3 = 2x^4 - \frac{x^4}{4}$$

$$\vdots$$

$$\ln(1-x) = -\frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} \cdots \infty$$

$$-\left(\ln(1-x) + x\right) = \left(\frac{x^2}{2} + \frac{x^3}{3} + \cdots \infty\right)$$

$$S_\infty = 2\left(x^2 + x^3 + x^4 + \cdots \infty\right) - \left(\frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \cdots \infty\right)$$

$$= \frac{2x^2}{1-x} + \left(\left(\ln(1-x) + x\right)\right)$$

TRIGONOMETRIC FUNCTIONS (RATIOS & IDENTITIES)

Question [JEE Mains-2013]

The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as:

83%

A sin A cos A + 1

B sec A cosec A + 1

C tan A + cot A

D sec A + cosec A

$$\frac{\sin^2 A}{\cos A(\sin A-\cos A)}-\frac{\cos^2 A}{\sin A(\sin A-\cos A)}$$

$$\frac{1}{(\sin A-\cos A)}\left\{\frac{\sin^2 A}{\cos A}-\frac{\cos^2 A}{\sin A}\right\}$$

$$\frac{1}{(\sin A-\cos A)}\cdot\frac{\sin^3 A-\cos^3 A}{\cos A\cdot\sin A}$$

$$(\sin A-\cos A)(\sin^2 A+\cos^2 A+\sin A\cos A)$$

$$=\frac{1+\sin A\cos A}{\sin A\cos A}$$

Challenger Question

Questions are based on elimination of θ :

(i) If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, then $(m^2n)^{2/3} + (mn^2)^{2/3} =$

(ii) If $a\sec\theta - c\tan\theta = d$ & $b\sec\theta + d\tan\theta = c$, then show $a^2 + b^2 = c^2 + d^2$

$a\sec\theta = d + c\tan\theta$ $\qquad$ $b\sec\theta = c - d\tan\theta$

SBS

$a^2\sec^2\theta = d^2 + c^2\tan^2\theta + 2dc\tan\theta$

$b^2\sec^2\theta = c^2 + d^2\tan^2\theta - 2dc\tan\theta$

add

$(a^2+b^2)\sec^2\theta = (d^2+c^2) + (d^2+c^2)\tan^2\theta$

$= (d^2+c^2)(1+\tan^2\theta)$

HHPP.



S.A.S.

$a\sin^2\theta + b\cos^2\theta = c$

$a(1-\cos^2\theta) + b\cos^2\theta = c$ ✓

$a\sin^2\theta + b(1-\sin^2\theta) = c$ ✓

$\div \cos^2\theta$

$a\dfrac{\sin^2\theta}{\cos^2\theta} + \dfrac{b\cos^2\theta}{\cos^2\theta} = \dfrac{c}{\cos^2\theta} = c\sec^2\theta$

$a\tan^2\theta + b = c(1+\tan^2\theta)$ ✓

$a\sin^2\theta + b\cos^2\theta = c\sin\theta\cos\theta$.

$\div \cos^2\theta$

$a\dfrac{\sin^2\theta}{\cos^2\theta} + \dfrac{b\cos^2\theta}{\cos^2\theta} = \dfrac{c\sin\theta\cos\theta}{\cos^2\theta}$.

$a\tan^2\theta + b = c\tan\theta$. ✓

Question [IIT-JEE-2009]

If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

M-1

A $\tan^2 x = \frac{2}{3}$

B $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

C $\tan^2 x = \frac{1}{3}$

D $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

$\cos^4 x = (\cos^2 x)^2 = (1 - \sin^2 x)^2$

$$\frac{\sin^4 x}{2} + \frac{1 + \sin^4 x - 2\sin^2 x}{3} = \frac{1}{5}$$

Say: $\sin^2 x = t$

$\frac{t^2}{2} + \frac{1 + t^2 - 2t}{3} = \frac{1}{5}$

$3t^2 + 2 + 2t^2 - 4t = \frac{6}{5}$

$5(5t^2 - 4t + 2) - 6 = 0$

$25t^2 - 20t + 4 = 0$

$25t^2 - 20t + 4 = 0$

$25t^2 - 10t - 10t + 4 = 0$

$5t(5t - 2) - 2(5t - 2) = 0$

$(5t - 2)^2 = 0$

$t = \frac{2}{5}$

Question [IIT-JEE-2009]

If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

$t = \frac{\sin^2 x}{\cos^2 x} = \frac{2/5}{3/5} \rightarrow 1 - \cos^2 x = \frac{2}{5}$

$= 1 - \frac{2}{5} = \cos^2 x$

$\# \tan^2 x = \frac{2}{3}$

$\sin^2 x = \frac{2}{5}$

$\cos^2 x = \frac{3}{5}$

$\frac{\sin^8 x}{8} = \left(\frac{\sin^2 x}{8}\right)^4 = \left(\frac{2}{5}\right)^4 = \frac{16}{8 \times 5^4} = \frac{2}{5^4}$

$\frac{(\cos^8 x)}{27} = \frac{3^4}{5^4 \times 3^3}$

$\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{5^4} + \frac{3}{5^4} = \frac{5}{5^4} = \frac{1}{5^3} = \frac{1}{125}$

* Cauchy Schwarz

$$(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2)$$

Titu's lemma:

$a_1 \to \frac{a_1}{\sqrt{b_1}}$ & $b_1 \to \sqrt{b_1}$ $\quad$ $a_2 \to \frac{a_2}{\sqrt{b_2}}$ & $b_2 \to \sqrt{b_2}$.

$$\left(\frac{a_1}{\sqrt{b_1}} \cdot \sqrt{b_1} + \frac{a_2}{\sqrt{b_2}} \cdot \sqrt{b_2}\right)^2 \leq \left(\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2}\right)(b_1 + b_2)$$

$$(a_1 + a_2)^2 \leq \left(\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2}\right)(b_1 + b_2)$$

Titu's lemma :

$$\left(\frac{x^2}{a}+\frac{y^2}{b}\right) \geq \frac{(x+y)^2}{a+b}$$

equality hold :-

$\frac{x}{a}=\frac{y}{b}$

ex :- If $x+y=3$ & min. value of $\left(\frac{x^2}{7}+\frac{y^2}{5}\right)$

$$\frac{x^2}{7}+\frac{y^2}{5} \geq \frac{(\overbrace{x+y}^{3})^2}{12}$$

$$\hookrightarrow \geq \frac{3}{4}$$

Question [IIT-JEE-2009]

If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$, then

(A) $\tan^2 x = \frac{2}{3}$

(B) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$

(C) $\tan^2 x = \frac{1}{3}$

(D) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

M-2

$$\frac{(\sin^2 x)^2}{2} + \frac{(\cos^2 x)^2}{3} = \frac{1}{5}$$

equality

$$\frac{\sin^2 x}{2} = \frac{\cos^2 x}{3}$$

$$\sec^2 x - 1 = \frac{2}{3}$$

$$\sec^2 x = \frac{5}{3}$$

$$\cos^2 x = \frac{3}{5}$$

$$\sin^2 x = \frac{2}{5}$$

$$\tan^2 x = \frac{2}{3}$$

$$\frac{(\sin^2 x)^2}{2} + \frac{(\cos^2 x)^2}{3} \geq \frac{(\overbrace{\sin^2 x + \cos^2 x}^{1})^2}{2+3}$$

$$\geq \frac{1}{5}$$

Question [JEE Mains-2021]

If $15\sin^4\alpha + 10\cos^4\alpha = 6$, for some $\alpha \in R$, then the value of $27\sec^6\alpha + 8\operatorname{cosec}^6\alpha$ is equal to:



A 500

B 400

C 250

D 350



BASIC OF TRIGONOMETRY (CLASS-X)

RAR

Point-03: Trigonometric Values

	0°	30°	45°	60°	90°
$\sin\theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	n.d.
$\cot\theta$	n.d.	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec\theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	n.d.
$\operatorname{cosec}\theta$	n.d.	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1 .

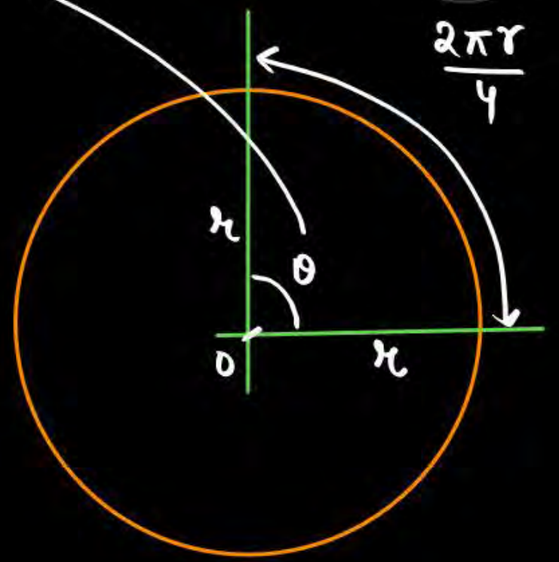
Degree

$90^\circ = \frac{\pi}{2}$

$180^\circ = \pi$

$1^\circ = \frac{\pi}{180}$ rad.

$\frac{180^\circ}{\pi} = 1$ rad.

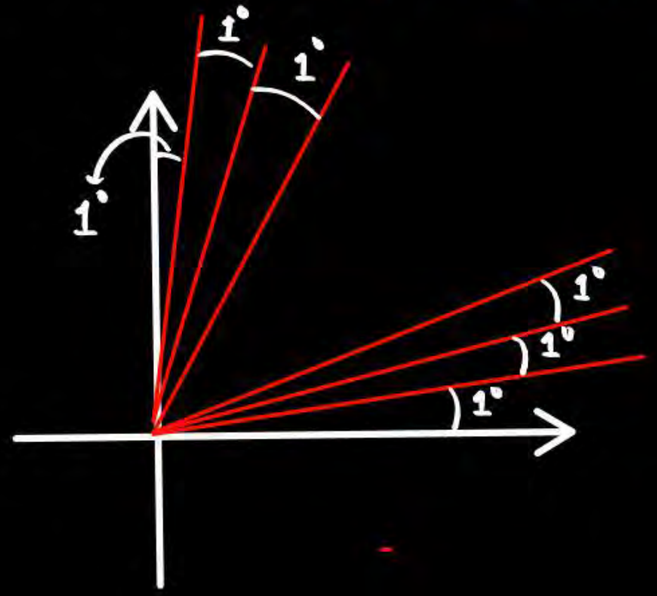


$r\theta = \frac{2\pi r}{4}$

$\theta = \frac{\pi}{2}$

ANGLES

Two Units: **Degree(Sexagesimal System)** **Radian(Circular System)**



Right angle ⇒ 90 equal parts

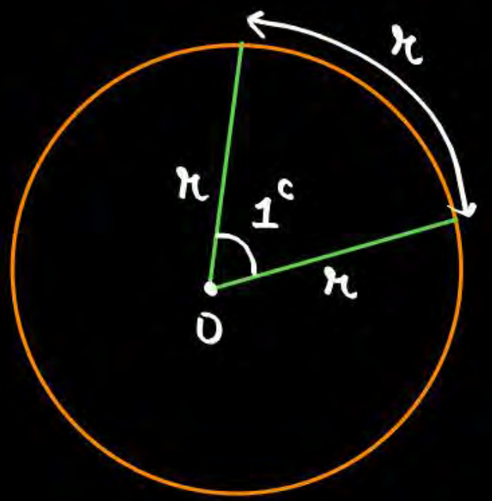
1° (each part)

1° divided into 60 equal parts

↳ each part is called as 1 minute (1′)

1′ div. into 60 equal parts

↳ each part is called as 1 second (1″)



$1^c = 1 = 1$ radian

degree°

↳ minute'

↳ second''

$1^\circ = 60'$

$1' = 60''$

IMPORTANT NOTE

(i) 1 radian $\approx (57.32)^\circ$

(ii) $\pi = 180^\circ$

(iii) $\theta(\text{in radian}) = \theta^c = \left(\frac{180}{\pi} \times \theta\right)^\circ$

(iv) θ (in degree) $= \theta^\circ = \left(\frac{\pi}{180} \times \theta\right)$ radian $= \left(\frac{\pi}{180} \times \theta\right)^c$

(v) Remember as Result (R.A.R.)

$\pi \simeq 3.14$

$\frac{\pi}{2} \simeq 1.57$

$\frac{\pi}{4} \simeq 0.78$

CONVERSIONS

$\frac{\pi}{2} = 90^\circ$

$\frac{\pi}{4} = 45^\circ$

$\frac{\pi}{3} = 60^\circ$

$\frac{\pi}{6} = 30^\circ$

$\frac{3\pi}{4} = 135^\circ$

$\frac{4\pi}{3} = 240^\circ$

$330^\circ = \frac{11\pi}{6}$

$225^\circ = \frac{5\pi}{4}$

$225 \times \frac{\pi}{180} = \frac{5\pi}{4}$

$270^\circ = \frac{3\pi}{2}$

$240^\circ = \frac{4\pi}{3}$

$360^\circ = 2\pi$

$12^\circ = \frac{\pi}{15}$

$12 \times \frac{\pi}{180} = \frac{\pi}{15}$

$24^\circ = \frac{2\pi}{15}$

$36^\circ = \frac{3\pi}{15} = \frac{\pi}{5}$

IMPORTANT NOTE

RAR (खून में बसा लो)

$30^\circ = \frac{\pi}{6}$	$45^\circ = \frac{\pi}{4}$	$60^\circ = \frac{\pi}{3}$
$90^\circ = \frac{\pi}{2}$	$120^\circ = \frac{2\pi}{3}$	$135^\circ = \frac{3\pi}{4}$
$150^\circ = \frac{5\pi}{6}$	$180^\circ = \pi$	$210^\circ = \frac{7\pi}{6}$
$225^\circ = \frac{5\pi}{4}$	$240^\circ = \frac{4\pi}{3}$	$270^\circ = \frac{3\pi}{2}$
$300^\circ = \frac{5\pi}{3}$	$315^\circ = \frac{7\pi}{4}$	$330^\circ = \frac{11\pi}{6}$

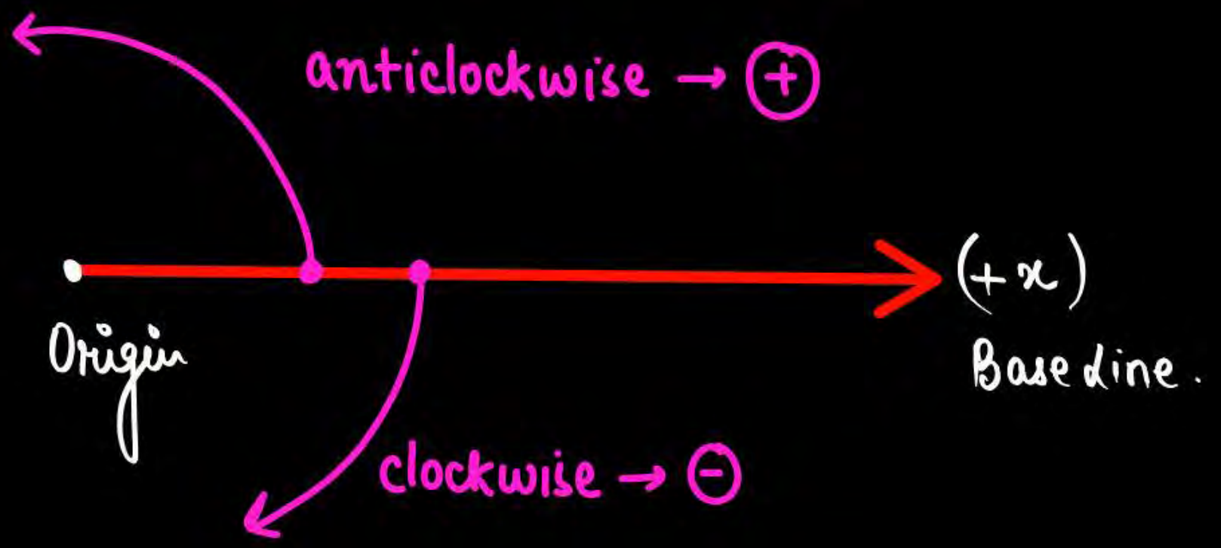
IMPORTANT NOTE

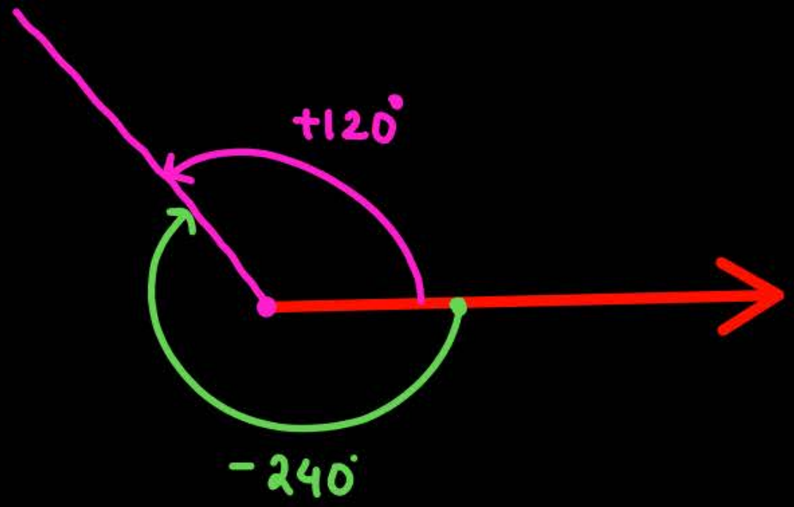
(1) $90°, 180°, 270°$ ... $\Rightarrow$ Allied angles

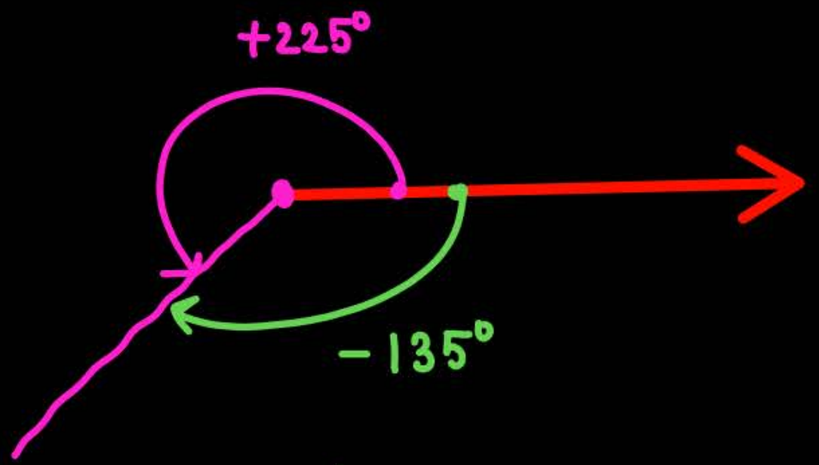
(2) If $\alpha + \beta = \pi \Rightarrow \alpha$ & β Supplementary angles

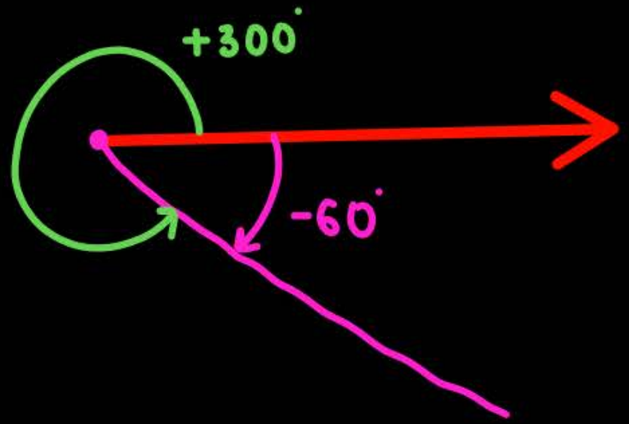
(3) If $\alpha + \beta = \frac{\pi}{2} \Rightarrow \alpha$ & β complementary angles

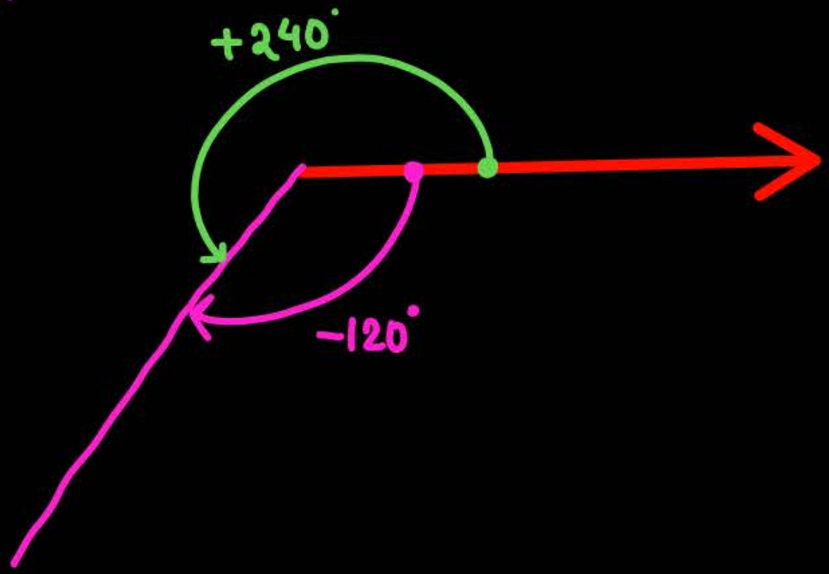
SIGN CONVENTION



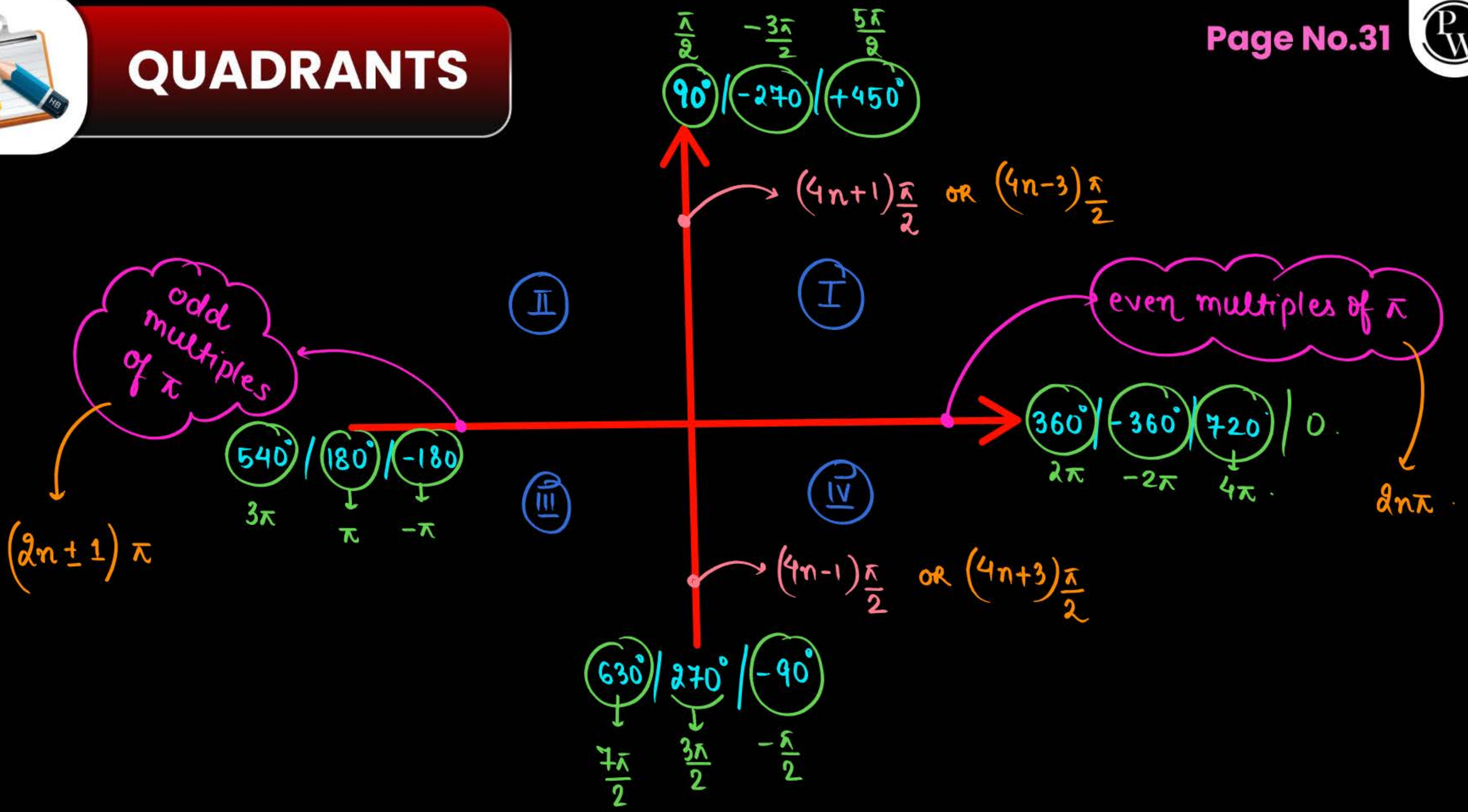

+120°
-240°


+225°
-135°


+300°
-60°


+240°
-120°

QUADRANTS


$\frac{\pi}{2}$ $-\frac{3\pi}{2}$ $\frac{5\pi}{2}$
90° | -270° | +450°
$(4n+1)\frac{\pi}{2}$ OR $(4n-3)\frac{\pi}{2}$
II
I
odd multiples of π
even multiples of π
540° | 180° | -180°
3π π -π
360° | -360° | 720° | 0.
2π -2π 4π
$(2n \pm 1)\pi$
$2n\pi$
III
IV
$(4n-1)\frac{\pi}{2}$ OR $(4n+3)\frac{\pi}{2}$
630° | 270° | -90°
$\frac{7\pi}{2}$ $\frac{3\pi}{2}$ $-\frac{\pi}{2}$

Ques :-

① $7\pi + \frac{\pi}{4} \longrightarrow$ III quad.

② $7\pi - \frac{\pi}{4} \longrightarrow$ II quad.

③ $1314572\pi + \frac{2\pi}{3} \longrightarrow$ II quad.

$\frac{2\pi}{3} \rightarrow 120°$

④ $-1314572\pi + \frac{\pi}{6} \longrightarrow$ I quad.

⑤ $\frac{7\pi}{2} - \frac{\pi}{4} \longrightarrow$ II quad.

⑥ $-\frac{7\pi}{2} - \frac{\pi}{4} \longrightarrow$ I quad.

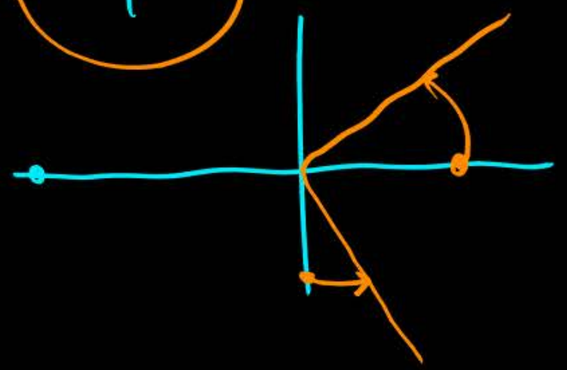
⑦ $151\frac{\pi}{2} + \frac{2\pi}{3} \longrightarrow$ I quad

$\frac{2\pi}{3} \rightarrow 120°$

$151 \rightarrow 4n+3$

(8) $\frac{151\pi}{3} = 50\pi + \frac{\pi}{3}$ → I quad.

$\frac{\cancel{150}^{50}\pi}{\cancel{3}} + \frac{\pi}{3}$



(9) $635^\circ = 90^\circ \times 7 + 5^\circ$ → IV quad.

Question

Identify Quadrants of following angles:

#DIBY !!

(i) 330°

(ii) -210°

(iii) -1120°

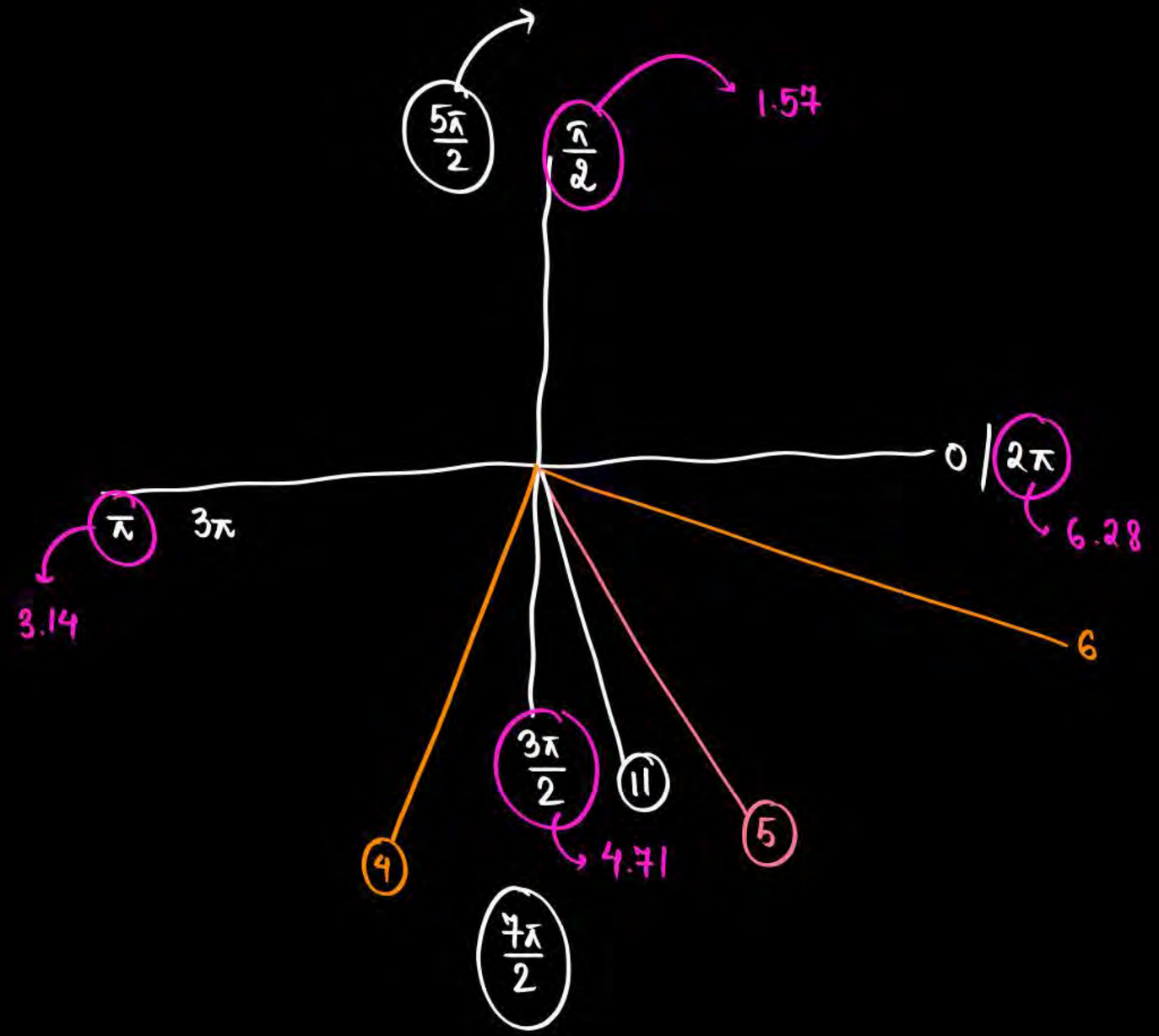
(iv) $\frac{11\pi}{5}$

(v) $\frac{57\pi}{4}$

(vi) $\frac{-31\pi}{6}$

(vii) $\boxed{5}$ IV quad.

(viii) $\boxed{11}$ IV quad.



(4) → III quad.

(6) → IV quad.

$\frac{22}{7} > \pi$

3.14258

3.141

$11 > \frac{7\pi}{2}$

DO IT BY YOURSELF (DIBY)

DIBY-01

If $\sin A \tan A = \cos^2 A$, then $\cos^3 A + \cos^2 A$ is equal to

(A) 1 (B) 2 (C) 4 (D) None of these

[Ans. : (A)]

DIBY-02

If $\text{cosec}A + \cot A = \frac{11}{2}$, then $\tan A$ is

(A) $\frac{21}{22}$ (B) $\frac{15}{16}$ (C) $\frac{44}{117}$ (D) $\frac{117}{43}$

[Ans. : (C)]

DIBY-03

If $\tan\theta = \frac{-4}{3}$, then $\sin\theta$ is

(A) $\frac{-4}{5}$ but not $\frac{4}{5}$ (B) $\frac{-4}{5}$ or $\frac{4}{5}$ (C) $\frac{4}{5}$ but not $-\frac{4}{5}$ (D) None of these

[Ans. : (B)]

DIBY-04

If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, then prove that $m^2 - n^2 = 4\sin\theta\tan\theta$

DIBY-05

Prove that $\frac{\tan A + \sec A - 1}{\tan A - \sec A + 1} = \frac{1 + \sin A}{\cos A}$

DIBY-06

Prove that: $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A\cos^2 A)$

DIBY-07

$(1 + \tan\alpha\tan\beta)^2 + (\tan\alpha - \tan\beta)^2 =$

(A) $\tan^2\alpha\tan^2\beta$

(B) $\sec^2\alpha\sec^2\beta$

(C) $\tan^2\alpha\cot^2\beta$

(D) $\sec^2\alpha\cos^2\beta$

[Ans. : (B)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu)- 4 sittings each of 1 HR

It is never about end result

It's always about Journey

Thank You

#futureIITians